

FISH 577: Applied Stock Assessment II (2-5 credits), Spring Quarter

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Prerequisites

Applied Stock Assessment I (FISH 576 for Winter 2027) or permission of instructor.

Overview

Applied Stock Assessment II is the second part of a two-quarter applied stock assessment series offered in collaboration with stock assessment scientists in the Fishery Resource Analysis and Monitoring Division at the Northwest Fisheries Science Center. This course focuses on developing, running, evaluating and documenting the Stock Synthesis (SS3) model required for a stock assessment update submitted to the Pacific Fishery Management Council (PFMC) for use in management. The course also continues to explore the Fisheries Integrated Modeling System (FIMS) model in parallel with SS3, to help with future transition to FIMS. During this second term, students will:

1. Run the update stock assessment model and complete a set of required model diagnostics.
2. Build decision tables to provide catch advice to managers.
3. Communicate results through a written report and presentations.

Students will focus on running the stock assessment model under different configurations and underlying assumptions, ensuring model convergence, interpreting and comparing results, and documenting the stock assessment update in an assessment report and presentations before the PFMC's Scientific and Statistical Committee's Groundfish Subcommittee. The PFMC's Scientific and Statistical Committee and its Groundfish subcommittee review the update stock assessment.

Format

One 1-hour lecture and one 1-hour laboratory each week in which the instructors first introduce the theory behind the topic being discussed and the practical approach including decision points and code. Depending on the topic, students start the process of applying the methods to the stock assessment update and possibly additional examples during the laboratory session.

Learning Goals

Upon successful completion of the course, students will be able to:

1. Read, understand and modify Stock Synthesis and FIMS input files.
2. Produce model results and plots from model outputs.
3. Run model diagnostics, such as likelihood profiles, sensitivity and retrospective analyses.
4. Develop assessment model forecasts, for use in management.
5. Write a detailed stock assessment report.
6. Present stock assessment results to the PFMC and potentially other stake holders.

Texts

There are no prescribed texts. Students will be assigned readings at the beginning of the class. Material is also covered in PowerPoint slides and example R scripts which will be posted on the class web-site.

Evaluation (CR/NC or Graded)

This is a CR/NC class. Students should sign up for a number of credits reflecting their expected involvement in the class and contributions to the projections and update stock assessment. Credit will depend upon participation in class and in the projections and update assessment. For variable credits, see requirements below:

2 credits: Attend and participate in class, read material, turn in homework assignments.

3 credits: Attend and participate in class, read material, turn in homework assignments. Participate in work up/write up of at least one data type/source for update stock assessment, run a sensitivity model and projection in Stock Synthesis and use R4SS to produce output figures.

4 credits: Attend and participate in class, read material, turn in homework assignments. Participate in work up/write up of at least one data type/source for update stock assessment, run a sensitivity model and projection in Stock Synthesis and use R4SS to produce output figures. Lead one section of the document preparation for data or sensitivity analyses

5 credits: Attend and participate in class, read material, turn in homework assignments. Serve as a lead on the update assessment, overseeing and reviewing each of the data analyses, sensitivity analysis, and document preparation. Develop decision tables for update assessment. Serve as a lead on development of presentation of update assessment.

Enrolling in the two and three credit versions of this course requires background in the R language for statistical computing, gained through course FISH 552, Introduction to R Programming for Natural Scientists or approval from the instructor.

Enrolling in the four and five credit versions of this course requires background in the R language for statistical computing, gained through courses FISH 552, Introduction to R Programming for Natural Scientists, and FISH 553, Advanced R Programming for Natural Scientists. Additional required course work includes FISH 555, Age-Structured Models in Fisheries Stock Assessment, and/or FISH 558, Decision Analysis in Natural Resource management, or approval from the instructor.

The expected hours of coursework per week for the credit hours options, as well as the breakdown of hours spent on course tasks are provided in the table below.

Credit Hours	Hours Per Week	Lecture	Laboratory	Other
2	6	1.5	1.5	3
3	9	1.5	1.5	6
4	12	1.5	1.5	9
5	15	1.5	1.5	12

Syllabus

Week	Topics
1	Review data topics from Applied Stock Assessment I. Documentation for update stock assessments. Overview of asar document preparation process. Overview of diagnostics to be complete.
2	Model bridging: Run a set of models transitioning from the previous assessment model to the current by updating one new piece of data at a time and running the model. Plot comparisons of the previous assessment outputs with each subsequent model runs until each dataset has been updated. Produce a full set of R4SS output for the fully updated model.
3	Model convergence diagnostics, jittering. Retrospective analysis: Complete 5-year retrospective runs. Generate assessment report template and start writing the stock assessment document.
4	Sensitivity analysis: Produce a set of model runs that include 1) model sensitivities from the last full (and any subsequent update) assessment, 2) any issues noted in the STAR or SSC reports, 3) runs that you are interested in completing, including any sensitivities that seem important given changes in parameter estimates in the update base model compared to the last model. Plot results against the base model. Complete Introduction, Data and Model Description sections of the assessment document.
5	Likelihood profiles: Complete likelihood profiles that were included in the last full assessment. Complete Model Diagnostics section of the assessment document.
6	Management history/changes in management. Risk neutrality and the science/management interface. Harvest Projections: Complete harvest projections and decision tables as included in the last full assessment. Complete Executive Summary of the assessment document.
7	Complete stock assessment document for instructors' review.
8	Complete stock assessment document for NMFS/PFMC internal review deadline.
9	Complete stock assessment document for PFMC briefing book deadline for the SSC review.
10	Prepare and practice presentation for the PFMC's SSC Groundfish Subcommittee review meeting.
Final week	Present update stock assessment to the PFMC's Scientific and Statistical Committee's Groundfish Subcommittee.

Religious accommodations

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at [Religious Accommodations Policy](https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/)

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Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form](https://registrar.washington.edu/students/religious-accommodations-request/) (<https://registrar.washington.edu/students/religious-accommodations-request/>).

Email and computer use

All students are expected to have an email address and you will receive email relevant to this course on a regular basis.

Academic Integrity

Trust between student and instructor is of paramount importance in academic settings. Plagiarism, cheating, and other misconduct are serious violations of the University of Washington Student Conduct Code (WAC 478-120) and your personal contract as a student. I expect that you will know and follow the university's policies on cheating and plagiarism. Please review the College of the Environment website on academic integrity so that you are clear on what constitutes academic misconduct. Any suspected cases of academic misconduct will be handled per University of Washington regulations. For more information, see the College of the Environment Academic Misconduct Policy and the University of Washington Community Standards and Student Conduct website. Be advised that as an instructor at the UW, I have the responsibility to notify University Conduct committees about any suspected student misconduct.

Disability Accommodations

It is crucial that all students in this class have access to the full range of learning experiences. At the University of Washington, it is the policy and practice to create inclusive and accessible learning environments consistent with federal and state law. Full participation in this course requires: 1) the ability to attend all lectures per week; 2) participate in small group discussions on topics relevant to the course, 3) participate in all laboratory/discussion sections and 3) prepare stock assessment documents and make short presentations that synthesize results and discussion orally. If you anticipate or experience barriers to your learning or full participation in this course based on a physical, learning, or mental health disability, please immediately contact the instructor to discuss possible accommodation(s). A more complete description of the disability policy of the College of the Environment can be found here. If you have, or think you have, a temporary or permanent disability that impacts your participation in any course, please also contact Disability Resources for Students (DRS) at: 206-543-8924 (V), 206-543-8925 (TDD), uwdss@uw.edu.